

Accuracy does not buy timing: a lead-time metric for stress anticipation in mobile and wearable sensing

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Abstract

Machine learning systems for stress detection are evaluated almost exclusively on whether they classify the current state correctly. They are rarely evaluated on whether they detect a stress transition early enough to support intervention — the property that matters most for deployment. We introduce ΔT , the signed time between a detected change point in a model’s **confidence predicted-stress** trajectory and the nearest ground-truth stress onset, and measure it on **two publicly available datasets**: StudentLife (smartphone behavioural sensing) and WESAD (physiological wearables). Our central finding is a null result with practical consequences. Across three model configurations whose micro-F1 spans 0.478 to 0.612 on StudentLife, the ΔT distributions are statistically indistinguishable: pairwise Wasserstein distances of 0.81–5.75 hours against a distribution width of approximately 0.611, timing does not differ detectably: among onsets matched in both models, paired mean ΔT differences are at most 2.2 hours and fall within a ± 6 -hour equivalence margin under two one-sided tests (in fact within ± 4.5 hours), and the median ΔT is zero for every configuration, against a spread of 28 to 30 hours, with all Mann-Whitney $p > 0.37$. The result survives a control that holds change-point segmentation granularity constant across models. Against two null models, none of the three shows significantly more anticipation than randomly timed change points produce under the same matching rule (observed rates 23–27%). Models retrained to forecast one to three EMA responses ahead (median gaps of 16 to 49 hours) lose no accuracy and keep a median ΔT of zero. Compared on identical validation sequences, their paired mean ΔT differs from the present-state models’ by at most 2 hours, with every interval including zero. A temporally asymmetric training objective

designed specifically to shift ΔT earlier also fails to move it, while incidentally improving macro-F1 by 1.27 to 2.05 points through class rebalancing. Four independent perturbations leave the anticipation window unchanged. Cross-modal extension to WESAD (66 matched pairs, anticipation rate 33.3% at penalty 0.5, stable at approximately 33% across three lower penalties) places the anticipation rate within a narrow band of 24.8–33.3% across two sensing modalities, two annotation protocols, and two dataset scales 3.1 to 3.4 points. We also report a methodological caution: the anticipation rate, a natural summary statistic, increases whenever can rise when the ΔT distribution widens regardless of whether it shifts earlier, producing a false positive without shifting earlier, and in our data it differs between models whose medians and paired mean differences do not. Change points are located retrospectively, so ΔT measures temporal alignment, not real-time warning lead. We conclude that timing and accuracy are separable properties of a stress-prediction model, that standard metrics are silent about the former, and that systems intended to anticipate rather than react will require methods that target timing directly.

Keywords: stress prediction, mobile sensing, anticipation, change-point detection, reproducibility, digital health, ecological momentary assessment

Tracked changes. Blue indicates additions; red indicates deletions. Tables and figures are shown in their revised form only.
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1. Introduction

Chronic and episodic stress contributes to a wide range of adverse health outcomes, including cardiovascular disease, immune dysfunction, sleep disruption, and elevated risk for mood and anxiety disorders (Cohen et al., 1983; McEwen, 2007; Chrousos, 2009). The public health burden of these downstream conditions has motivated a decade of work on stress detection from continuous mobile and wearable sensing (Wang et al., 2014; Schmidt et al., 2018; Sano and Picard, 2013; Gjoreski et al., 2017). The underlying premise of that work is preventive: if a system can identify stress early, intervention becomes possible before consolidation into a chronic pattern or a physiological episode.

Predictive healthcare, as distinguished from reactive diagnosis, depends on lead time. A model that identifies deterioration only after symptomatic onset is useful for retrospective analysis; it is not useful for preventive action. In cardiac monitoring, in glycaemic prediction, and in early-warning scoring

for sepsis (Henry et al., 2015; Islam et al., 2020; Goldstein et al., 2017)(Ebrahimzadeh et al., 2018; De Bois et al., 2022; Henry et al., 2015; Goldstein et al., 2017), the interval between model detection and clinical event is treated as a core evaluation quantity alongside sensitivity and specificity. In stress prediction, no such quantity exists.

The absence is consequential. Standard machine learning metrics — micro-F1, macro-F1, accuracy, area under the ROC curve — are all computed relative to per-window labels. A model that consistently recognises stress six hours after the person felt it and a model that recognises it an hour before receive identical scores on all of them. For a system whose stated purpose is to enable a timely response — suggesting a break, prompting a coping exercise, escalating to a clinician — the two are not equivalent.

This paper introduces a measurement for the missing quantity, applies it to two datasets from complementary sensing modalities the StudentLife smartphone-sensing dataset, and reports what it shows. Models trained to predict the current label are not, of course, designed to anticipate. The point of ΔT is to make the timing property measurable rather than assumed — and then to test directly whether it responds to the levers available to a system designer: higher accuracy, personalisation, a timing-targeted objective, or training the model to forecast ahead. ΔT is defined as the signed time between a change point in the model’s own confidence stress trajectory and the nearest ground-truth stress onset. Negative values indicate the model detected the transition before the person reported it — anticipation. Positive values indicate reactive detection.

The framework is deliberately model-agnostic. It requires only that the model emit class probabilities over a temporally ordered evaluation set, which every modern classifier does by default. It requires no changes to model architecture, no additional annotation, and no retraining. It sits alongside standard classification metrics as a complementary measurement of lead time.

Our contributions to predictive healthcare research are as follows.

1. **A lead-time metric for stress anticipation.** ΔT formalises the notion of anticipation in a form directly comparable to lead-time metrics used in other predictive-healthcare domains such as cardiac arrhythmia forecasting, glycaemic prediction, and sepsis early warning.
2. **An invariance equivalence result across architectures.** Across three model configurations spanning 13.4 13.3 points of micro-F1 on StudentLife, the ΔT distributions are statistically indistinguishable. The result survives a control for segmentation sensitivity.

3. **Cross-modal extension to WESAD.** Applying the same framework to a physiological wearable dataset (electrocardiogram, electrodermal activity, blood volume pulse, and related channels) produces anticipation rates within a narrow band of 24.8–33.3% across two sensing modalities, two annotation protocols, and two dataset scales. The measurement is stable across substantially different sensing regimes. **paired mean ΔT differences on onsets matched in both models fall within a ± 6 -hour equivalence margin, with inference from a bootstrap that resamples students. No configuration shows significantly more anticipation than either of two chance-level null models, and the pattern persists under a control for segmentation sensitivity.**
4. **A negative result Negative results on an two anticipation-targeted objective interventions.** A temporally asymmetric loss function constructed specifically to shift ΔT earlier does not shift it. **It produces , producing instead an unrelated and genuine improvement in class balance. This is gain in macro-F1. Nor does retraining the models to forecast one to three EMA responses ahead: classification accuracy is unchanged, and on identical validation sequences the forecast-trained and present-state models' paired mean ΔT differ by at most 2 hours. Both are informative for the design of future intervention-oriented objectives systems.**
5. **A methodological caution.** The anticipation rate, a natural summary statistic, **increases mechanically whenever can increase when the ΔT distribution widens . Reporting it alone produces a false positive in our data without shifting. In our data it differs between models whose medians and paired mean differences do not.** We recommend that predictive-healthcare studies report the median, interquartile range, and a distributional distance alongside the anticipation rate.

We also report an independent reproduction of the published baselines we build on. That reproduction is a precondition for the rest of the analysis rather than a contribution in itself, and we treat it accordingly.

The findings have direct implications for the design and evaluation of predictive-healthcare systems built on mobile and wearable sensing. If lead time is fixed by the sensing modality and annotation protocol rather than by the model, investment in classifier accuracy will not by itself produce more clinically actionable systems. Progress will require either (a) methods that explicitly target the timing property, or (b) modifications to sensing and annotation infrastructure to shorten the ground-truth delay itself.

2. Related work

2.1. Stress detection from mobile and wearable sensing

Passively collected smartphone signals predict self-reported stress meaningfully above chance. Early work by Sano and Picard (2013) demonstrated the feasibility of continuous stress recognition using accelerometer, skin conductance, and phone usage data. Wang et al. (2014) extended this to a ten-week longitudinal benchmark on 48 undergraduates at Dartmouth College, combining phone sensing with daily ecological momentary assessment, and the resulting dataset remains one of the most widely used public resources for naturalistic stress and mental-health sensing. On the physiological side, Schmidt et al. (2018) introduced WESAD, a controlled multimodal wearable dataset collected under the Trier Social Stress Test protocol, spanning electrocardiogram, electrodermal activity, blood volume pulse, electromyogram, skin temperature, respiration, and three-axis acceleration from chest and wrist devices.

Across this literature, modelling individual differences consistently improves performance, and personalisation schemes that condition on subject identity or subject-level characteristics are common. Deep learning approaches, particularly LSTM autoencoders and their personalised extensions, have demonstrated strong results on these tasks (Information Fusion Lab, 2021; Hovsepian et al., 2015; Taylor et al., 2020). Complementary work using galvanic skin response and other physiological signals has established the feasibility of stress pattern detection from wearable sensors alone (Bakker et al., 2011; Gjoreski et al., 2017).

Instrumented self-report of stress is the standard ground truth in this literature, with the Perceived Stress Scale (Cohen et al., 1983) and daily ecological momentary assessment prompts the two most common protocols. Both introduce delay between the underlying physiological or behavioural signal and the recorded label; this delay is one of the properties that a lead-time metric such as ΔT implicitly measures.

2.2. Predictive healthcare and lead-time metrics

Predictive-healthcare systems in adjacent domains are routinely evaluated on how far in advance they detect events. Cardiac arrhythmia predictors are characterised by warning windows before atrial fibrillation episodes (Islam et al., 2020) Paroxysmal atrial fibrillation predictors are evaluated on their ability to flag the interval immediately preceding an episode (Ebrahimzadeh et al., 2018). Sepsis early-warning scoring

systems are evaluated on the interval between prediction and clinical onset (Henry et al., 2015; Goldstein et al., 2017). Continuous glucose monitoring combined with machine learning **uses is benchmarked at explicit** prediction horizons of **15 to 60 minutes30 to 120 minutes**, to enable preemptive dosing decisions (De Bois et al., 2022). Across these domains, lead time is measured directly, not inferred from classification metrics.

Stress prediction has not adopted this practice. The absence is sometimes attributed to the inherent difficulty of establishing a precise onset timestamp for a self-reported subjective state. Change-point detection provides an operational answer: rather than require an externally validated onset, define the onset as the moment the recorded label crosses a threshold and treat the model’s own **confidence stress** trajectory as an independent observation whose alignment with that moment can be measured. This is the approach we adopt.

2.3. Temporal structure in health sensing

Most work treats stress prediction as static classification: given a window of sensor data, predict the label for that window. The temporal dynamics, and in particular whether a model’s internal state shifts before the corresponding shift appears in self-report, have received considerably less attention. The notion of an anticipation window is implicit in work on affect forecasting and in the predictive-maintenance literature more broadly (Lei et al., 2018), but to our knowledge has not been formalised as a directly measurable quantity for stress prediction prior to this work.

2.4. Change-point detection

Change-point detection has an established history in physiological and behavioural time series. The PELT algorithm (Killick et al., 2012) provides an exact solution with linear expected computational cost under a broad family of cost functions, which suits the relatively short per-subject sequences typical of ecological momentary assessment studies. Related methods based on dynamic programming (Jackson et al., 2005) and a broader family of offline detection procedures are reviewed by Truong et al. (2020). Our use of these methods differs from the standard application: we apply change-point detection to the model’s output trajectory rather than to the input signal, converting a detection tool into an instrument for measuring model behaviour.

2.5. Reproducibility in machine learning for health

Reproducibility difficulties are widely documented across machine learning (Pineau et al., 2021). Health applications face additional obstacles from restricted data, privacy constraints, and the distance between research prototypes and deployable systems (Beam and Kohane, 2018; Futoma et al., 2020). Our own experience during this work reinforces a narrower point: configuration errors that produce plausible-looking results are a distinct and under-discussed failure mode, separate from the more commonly cited issues of hyperparameter sensitivity and random seed variation. We document this in Section 4.

3. Data and models

3.1. Dataset

StudentLife (Wang et al., 2014) was conducted at Dartmouth College over a ten-week academic term in spring 2012. Forty-eight undergraduate students carried an instrumented Android phone that logged GPS traces, Wi-Fi access points, accelerometer activity, ambient audio features, call and message metadata, and application usage. Daily ecological momentary assessment prompts collected self-reported stress on a five-point scale, collapsed for this task into three classes: no or minimal stress, mild stress, and high stress.

After the preprocessing supplied with the reference implementation, 1,175 labelled examples remain across 23 students, each with 14 normalised feature channels. The class distribution is 22.1%, 43.4%, and 34.5% respectively; this imbalance depresses macro-averaged F1 relative to micro-averaged F1 throughout.

Each example carries a compound key encoding subject identity and observation time at hour resolution. This permits reconstruction of a per-subject timeline directly from the evaluation set, without auxiliary timestamp files, which is what makes the ΔT computation possible on this dataset.

WESAD (Schmidt et al., 2018) is a controlled laboratory dataset with 15 subjects wearing chest and wrist sensors during the Trier Social Stress Test. We reproduced the reference implementation’s WESAD results, obtaining micro-F1 = 0.8944 ± 0.126 under leave-one-subject-out cross-validation after correcting three data-handling problems: a reconstructed covariate file (1 KB) was replaced by the authors’ 29.7 MB public release asset; label-distribution information was removed from subject portrait vectors (target leakage); and PCA dimensionality reduction was refitted using only training subjects (structural leakage).

The pipeline persists softmax probability arrays alongside true class labels, allowing direct ΔT extraction without retraining.

3.2. Model configurations

We use the reference implementation released by the Information Fusion Laboratory at the University of Massachusetts Amherst (Information Fusion Lab, 2021). Three configurations are compared:

- **LSTM baseline.** A multitask LSTM autoencoder without personalisation. The reference implementation’s lstm configuration: the LSTM autoencoder architecture without personalisation, trained with the classification loss only (the reconstruction term is switched off in this configuration, so the decoder is not trained). All subjects share a single output head.
- **Personalised, one branch.** Per-subject output heads with The calm_net configuration: the same LSTM autoencoder trained with the reconstruction term, per-subject output heads, and a single-branch routing module.
- **Personalised, three branches.** Per-subject output heads As above with a three-branch routing module.

These are the configurations the reference implementation designates as its baseline and its personalised models. They differ in two respects, personalisation and the reconstruction term, so the accuracy gap between them should not be attributed to personalisation alone. For this paper what matters is that they are models of clearly different accuracy trained on the same folds.

Personalisation operates through a subject-to-group mapping combined with per-subject output layers held in a module dictionary keyed by subject identifier. The architecture used in the WESAD pipeline is a convolutional-embedding transformer using the tAPE positional encoding scheme of Foumani et al. (2024), rather than an LSTM; the two datasets use distinct backbone architectures.

3.3. Training protocol

Ten repetitions of five-fold cross-validation, giving 50 folds in total, with 200 epochs per fold. Folds are constructed by student group rather than by example. This distinction is material: splitting by example would place the same subject in both training and validation partitions, inflating those of the reference implementation: a deterministic stratified five-fold split at the example level, stratified jointly by student and stress label.

Every student therefore contributes examples to both the training and the validation partition of every fold. This is what makes per-subject output heads usable at validation time, and it means the reported F1 for reasons unrelated to generalisation. Splitting by subject requires the model to face individuals it has not seen measures within-subject generalisation to unseen days, not generalisation to unseen individuals. Each student’s validation examples in a fold form a temporally sparse subsequence of that student’s timeline (about one fifth of their EMA responses), and it is on this subsequence that ΔT is computed. The split depends only on the labels, so all configurations trained on the same data file share identical folds; this is what makes the onset-paired comparisons of Section 6.3 possible.

We report the Following the reference implementation, each fold is summarised by its peak validation micro-F1 per fold , matching over the 200 epochs, and the model outputs at that epoch are the ones used for the ΔT analysis. This rule is applied identically to every configuration, including the asymmetric and forecasting runs. Because the epoch is chosen on the validation fold itself and there is no separate test set, the reported F1 values are optimistic; we use them to rank configurations, as the reference implementation ’s checkpointing behaviour. Training was conducted on an NVIDIA does, and not as estimates of deployed accuracy. All models were retrained for this revision on NVIDIA A30 and H100 80 GB GPU, requiring approximately 9 minutes per foldGPUs, one repetition (five folds) per job, with a distinct random seed per repetition.

4. Reproduction

Both published baselines reproduce. Table 1 and Figure 1 report the results.

Table 1: Reproduction of published baselines on StudentLife. Mean and standard deviation of micro-F1 across 50 folds (10 repetitions of 5-fold cross-validation), except the authors’ released checkpoint, which contains 5 folds. All configurations were retrained for this revision; every result in Section 6 derives from these runs.

Configuration	Micro-F1	Published target
LSTM baseline	0.4783 ± 0.030	≈ 0.47
Authors’ released checkpoint	0.5991	≈ 0.60
Personalised, one branch	0.6034 ± 0.028	≈ 0.60
Personalised, three branches	0.6112 ± 0.022	≈ 0.60

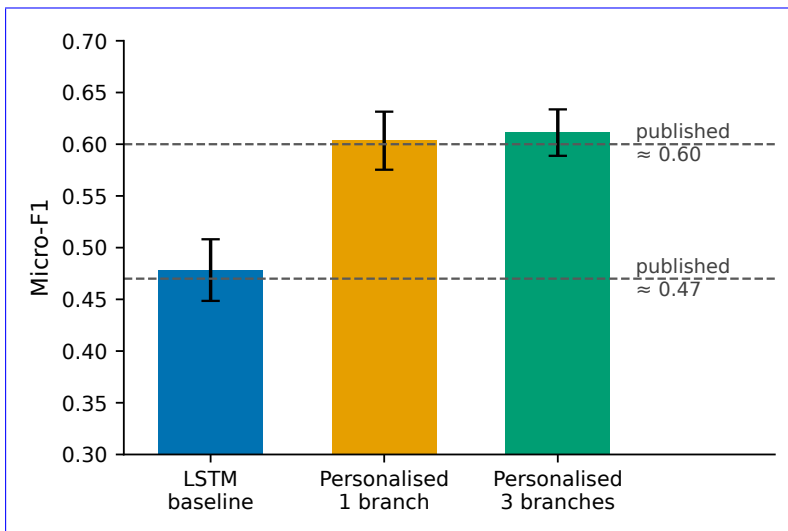


Figure 1: Reproduction of published baselines. Error bars show one standard deviation across 50 folds. Dashed lines mark the published targets. Both targets are met.

Four independent lines of evidence confirm the personalised result: the authors’ released checkpoint (0.5991), two independently trained configurations (0.6028 and 0.6115, 0.6034 and 0.6112), and the originally published figure.

We note one methodological point that arose during this reproduction. An earlier phase of this work concluded, incorrectly, that the personalised model failed to reproduce its published performance. The cause was a configuration error: the training entry point accepts six positional arguments, of which the fifth selects the subject-to-group mapping and the sixth is an unused output label. Three supposedly distinct configurations had been run with the same value for the fifth argument, varying only the sixth. They were the same model executed three times, and the near-identical results they produced were interpreted as a substantive finding.

This failure mode deserves documentation because the resulting numbers were entirely plausible. The fifth argument controls whether the model uses a mapping that assigns all 23 subjects to a single group (`all_in_one.pkl`, 718 bytes) or a mapping that assigns each subject to its own group (`one_for_each.pkl`, 731 bytes). Both files shipped with the repository. Personalisation is meaningless under `all_in_one`: every subject shares one output head. All results in this paper were produced after identifying and correcting this error.

5. The ΔT metric

We define the anticipation window as:

$$\Delta T = t_{\text{model}} - t_{\text{self-report}} \quad (1)$$

where t_{model} is the timestamp of a change point in the model’s **confidence stress** trajectory and $t_{\text{self-report}}$ is the timestamp of the nearest ground-truth stress onset within a matching window. Negative ΔT indicates anticipation; positive ΔT indicates reactive detection. **The change points are located offline, using each subject’s complete validation trajectory, so ΔT measures retrospective temporal alignment between the model’s trajectory and the reported onsets. It is not the lead time of an alert that a deployed system could have issued in real time; we return to this in Section 7.3.**

5.1. Pipeline

Fold assignments used during training are recovered by invoking the reference implementation’s split function with identical parameters and a fixed seed

***Timeline reconstruction.* The identifiers of each fold’s validation examples are stored with the fold’s training record, so the analysis reads them directly instead of reconstructing the split. Each validation example’s compound key parses to a timestamp expressed as hours since a fixed epoch.**

***Stress trajectory.* For each example we compute the per-row softmax $\mathbf{p}_t = (p_t^{(0)}, p_t^{(1)}, p_t^{(2)})$ over the three class logits and take the maximum class probability as a scalar confidence value ordered stress classes and form the expected stress level**

$$s_t = \sum_{c=0}^2 c p_t^{(c)} \in [0, 2], \quad (2)$$

which increases monotonically with the probability mass placed on higher stress and therefore represents stress directly. Examples are grouped by subject and sorted chronologically. The originally submitted analysis used the maximum class probability $\max_c p_t^{(c)}$, which measures model certainty rather than stress; we retain it only as a sensitivity analysis (Appendix A).

Change-point detection. We apply PELT (Killick et al., 2012) with a radial basis function cost to each subject’s **confidence trajectory** **stress trajectory**, using a **minimum segment length of two examples and a candidate grid of every example ($\text{jump} = 1$)**. The penalty was calibrated by scanning $\{0.5, 1.0, 2.0, 5.0\}$ on a representative fold; a penalty of 1.0 was selected as neither over- nor under-segmenting.

Matching. Ground-truth stress onsets are upward transitions in the chronological label sequence. Each detected change point is matched to its nearest onset within a 72-hour window, and the signed difference recorded. Subjects contributing fewer than five examples to a fold are excluded.

5.2. An implementation detail worth recording

The **ruptures** library (Truong et al., 2020) returns a list of change-point indices whose final element equals the length of the signal, functioning as a terminating sentinel rather than a detected change. Retaining that element introduces one spurious change point per subject, positioned at the end of the series and therefore systematically late.

In our data this single error shifted the measured anticipation rate from 29.3% to 12.8%. **A second library default deserves the same warning: ruptures evaluates candidate change points only on a grid of every fifth sample unless jump is set explicitly.** On StudentLife validation subsequences of roughly ten examples per subject, the default leaves only one or two admissible positions. The originally submitted analysis used this default; all results in this revision use **jump = 1**, and Appendix A reports the submitted configuration for comparison: the coarse grid admits roughly a third as many matched pairs (332–420 against 914–1120) and moves the median ΔT from 0 to between +4.0 and +5.5 hours. We report it because the resulting figure was plausible and, taken at face value, supported a substantive but incorrect conclusion.

6. Results

6.1. ΔT under a fixed penalty

Table 2 and Figures 2 and 3 report ΔT for the three configurations.

Pairwise comparisons on the pooled ΔT values give Wasserstein distances of 3.55, 5.75, and 2.64 hours **As a descriptive summary, the pairwise Wasserstein distances between the pooled ΔT distributions are 2.65, 2.45, and 0.85 hours (baseline–1-branch, baseline–3-branch, 1-branch–3-branch), with Mann-Whitney p values of 0.507, 0.908, and 0.559**

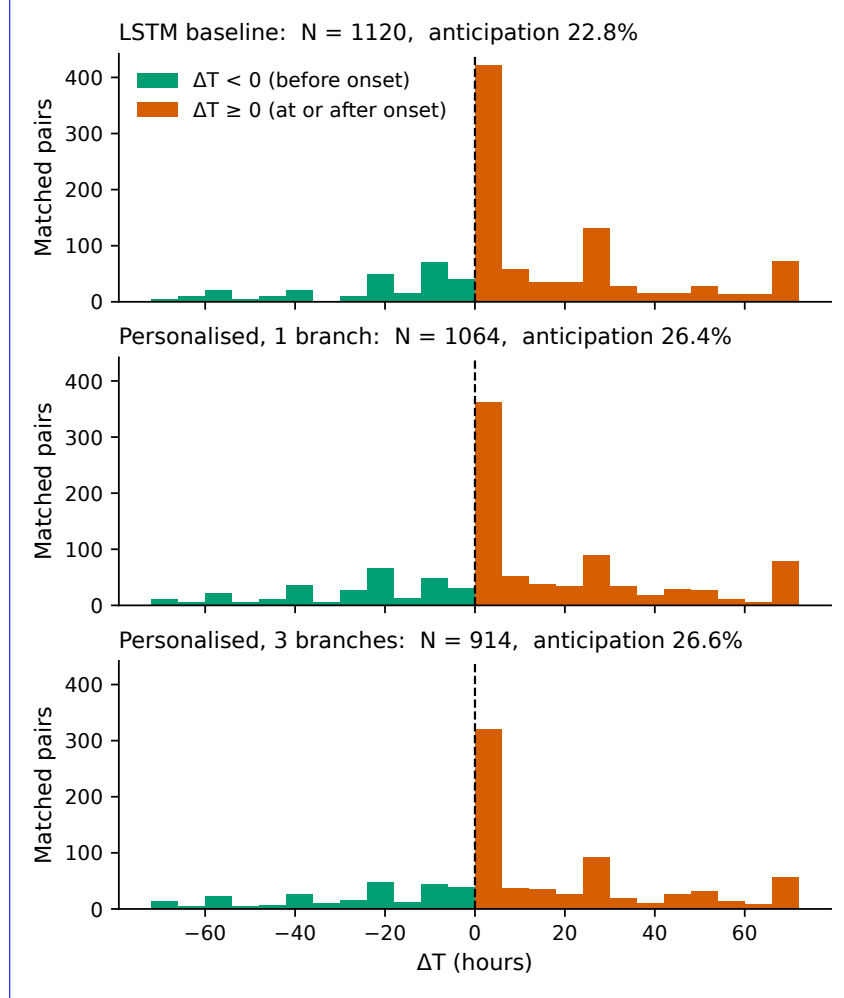


Figure 2: ΔT distributions for the three configurations. Green bars indicate anticipated transitions ($\Delta T < 0$); orange bars indicate reactive detections ($\Delta T \geq 0$). Dashed vertical lines mark the median (0 hours in all three panels). Panel annotations are generated from the same records as Table 2. The three panels are visually near-identical despite a 13.3-point spread in micro-F1.

Table 2: ΔT statistics under PELT with penalty 1.0, pooled over ten repetitions per configuration. The anticipation rate is the proportion of matched pairs with $\Delta T < 0$, reported pooled and as mean $\pm$ standard deviation across the ten repetitions; ties ($\Delta T = 0$, also shown) count as reactive. All time values are in hours.

Configuration	F1	N	Mean	Median	Std	IQR	$\Delta T = 0$	Anticipation
LSTM baseline	0.478	1120	+8.40	0	27.8	24.0	35.3%	22.8% (22.8 ± 2.1)
Personalised, 1 br.	0.603	1064	+6.97	0	30.4	30.0	32.8%	26.4% (26.4 ± 2.7)
Personalised, 3 br.	0.611	914	+6.65	0	30.0	29.0	32.6%	26.6% (26.7 ± 4.8)

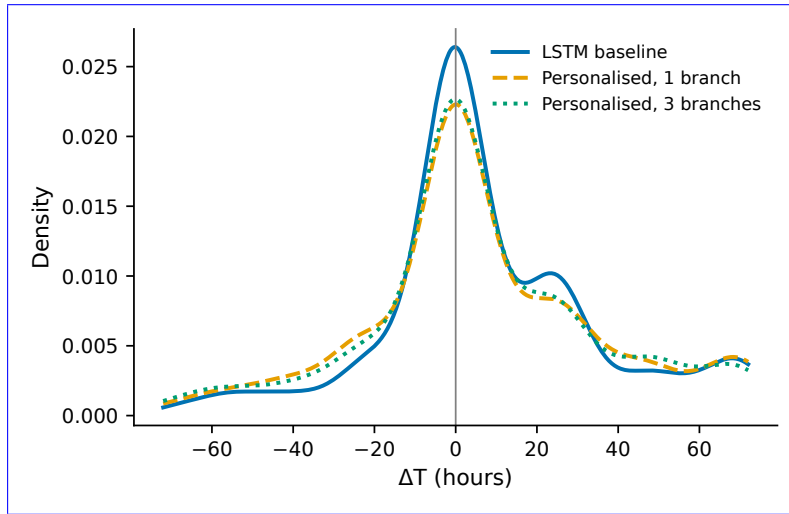


Figure 3: Kernel density estimates of the three ΔT distributions. The curves nearly coincide despite a 13.3-point spread in micro-F1.

respectively (Figure 4) 0.217, 0.188, and 0.917. Because matched pairs that share an onset or a subject are not independent, we do not treat these pooled tests as inferential; the paired analysis of Section 6.3 is the inferential comparison. We note also that the sample sizes differ across configurations (1120, 1064, 914), indicating that the fixed penalty segments each model’s stress trajectory at a different effective sensitivity; Section 6.4 removes this confound.

6.2. Null models: what chance timing produces

A ΔT distribution is only interpretable against what the same matching rule yields when the change points carry no information about the onsets. We constructed two null models, each with 1,000 draws per configuration.

Random change points. Within each subject-fold trajectory, the detected change points are replaced by the same number of positions drawn uniformly at random, and the matching is repeated.

Shifted labels. Each subject’s label sequence is circularly shifted by a random offset, preserving its autocorrelation and onset count while breaking any alignment with the model trajectory; the model’s actual change points are then matched against the shifted onsets.

We compare three statistics: the anticipation rate, the median ΔT , and an alignment statistic — the proportion of change points landing within ± 24 hours of an onset. Table 3 and Figure 4 report the results.

Table 3: Observed statistics against the two null models (mean and 95% interval over 1,000 draws). p_{ant} is the proportion of null draws with an anticipation rate at or below the observed one; p_{align} the proportion with alignment at or above the observed one. No configuration’s anticipation rate is significantly above either null; alignment is above the random-change-point null only for the baseline, and no configuration is above the shifted-label null.

		Baseline	Pers., 1 br.	Pers., 3 br.
Observed	Anticipation (%)	22.8	26.4	26.6
	Within ± 24 h of onset (%)	51.0	46.1	47.4
Random change points	Anticipation (%)	26.2 [23.8, 28.6]	25.9 [23.6, 28.1]	25.9 [23.5, 28.4]
	p_{ant}	< 0.01	0.67	0.70
	Within ± 24 h (%)	46.5 [44.5, 48.5]	46.2 [44.1, 48.2]	45.5 [43.2, 47.9]
	p_{align}	< 0.01	0.57	0.06
Shifted labels	Anticipation (%)	40.0 [37.6, 42.3]	39.7 [37.0, 42.3]	38.5 [35.8, 41.4]
	p_{ant}	< 0.01	< 0.01	< 0.01
	Within ± 24 h (%)	49.6 [47.8, 51.4]	51.2 [49.2, 53.1]	50.3 [48.0, 52.3]
	p_{align}	0.07	1.00	1.00

Two observations

Three facts follow. First, ΔT changes little across models whose micro-F1 differs by 13.4 points; no configuration shows significantly more anticipation than randomly timed change points produce under the same matching rule. The personalised models’ rates (26.4% and 26.6%) sit just above the random-change-point mean of 25.9% and well inside its 95% interval (67% and 70% of null draws fall at or below them); the baseline’s rate (22.8%) is significantly *below* it. An anticipation rate near 26% is therefore what this matching rule yields without any timing information, and is not by itself evidence of early warning. Second, the sample sizes differ across configurations (319, 458, 354), indicating the fixed penalty segments against the shifted-label null, whose anticipation rate is 38–40%, every observed rate is far

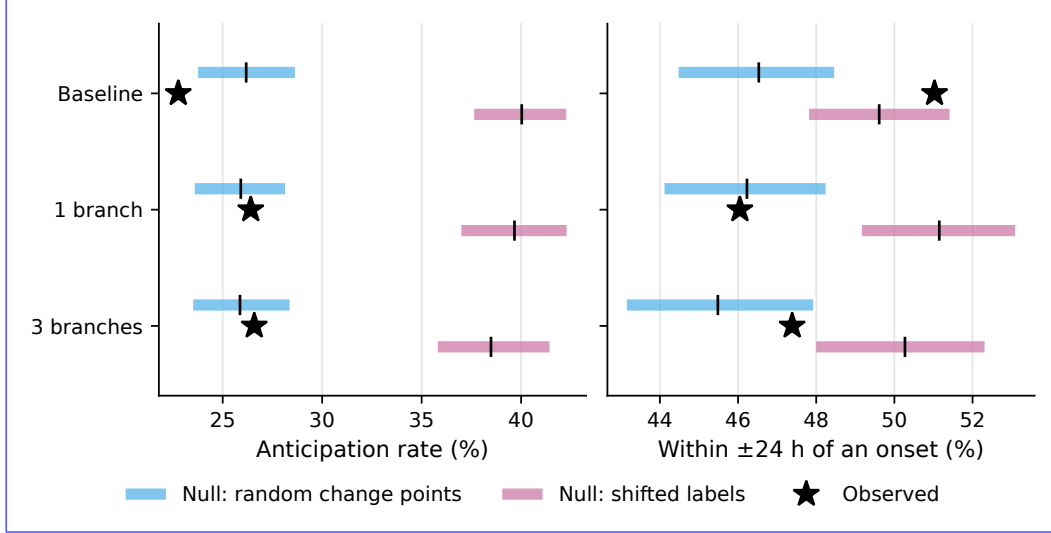


Figure 4: Observed anticipation rate (left) and onset alignment (right) against the two null models. Stars are the observed values; bars are the 95% intervals of each null over 1,000 draws, with the null mean marked. No observed anticipation rate is significantly above either null.

lower ($p < 0.01$): relative to onsets placed at unrelated positions, the models' change points fall before an onset much less often, which is the pattern expected of detection at or after the onset. Third, the alignment statistic gives a mixed picture. The baseline's change points fall within ± 24 hours of an onset more often than random change points do (51.0% against 46.5%, $p < 0.01$), but not more often than under shifted labels ($p = 0.07$); neither personalised model is above either null. The two nulls give different reference values for the same statistic (26% and 39% anticipation), which is itself a reason not to interpret an anticipation rate without stating the reference. In summary, no configuration showed significantly greater anticipation than either null; the baseline showed greater onset alignment than the random-change-point null, but not than the shifted-label null.

6.3. Onset-paired equivalence between models

Because every configuration shares identical folds (Section 3.3), each ground-truth onset can be matched across models. For each onset we take the ΔT of each model's confidence distribution at a different effective sensitivity. The comparison is not fully controlled nearest change point within the 72-hour window (median across the ten repetitions) and compare models on the paired differences. An onset enters a

pair only if both models have a change point within the window in at least one repetition; onsets matched in one model but not the other are excluded from that pair, so the analysis concerns onsets that both models respond to.

All inference is from a bootstrap that resamples *students* with replacement (2,000 draws) and recomputes the paired mean difference, so that multiple onsets from one student, and the ten repetitions, are not treated as independent. The confidence intervals are bootstrap percentile intervals. For equivalence we use a margin of ± 6 hours, a quarter of the daily EMA cycle and the smallest advance we consider actionable for a same-day intervention. The margin was fixed when the revision analysis was designed, before the models were retrained and before any result in this section was computed. The two one-sided tests (TOST) are carried out on the same student-level bootstrap distribution: the TOST p value is the larger of the proportions of bootstrap means at or beyond each margin, and equivalence at the 5% level corresponds to the 90% interval lying inside the margin. Because a reader may prefer a different margin, we also report the smallest margin at which equivalence holds. We additionally report an onset-level Wilcoxon signed-rank p value as a description only: it treats onsets as independent and should not be read as an inferential result.

The tested quantity is the *mean* paired difference. The analysis does not establish that the ΔT distributions have the same shape, tails, or anticipation rate; those are described in Sections 6.1 and 6.5.

Table 4: Onset-paired comparisons of ΔT between models. The paired difference is row model minus column model, in hours, with 95% student-clustered bootstrap confidence intervals. Equivalence is assessed by TOST at the pre-specified ± 6 -hour margin; the final column gives the smallest margin at which equivalence holds (the larger absolute bound of the 90% CI). The TOST p value and the intervals come from the student-level bootstrap; the Wilcoxon p value is onset-level and descriptive.

Pair	Onsets	Mean diff. [95% CI]	Wilcoxon p	TOST p	Equivalent at ± 6 h	Smallest margin
Baseline vs. 1 br.	163	+2.16 [−0.88, +5.10]	0.509	0.005	yes	4.5 h
Baseline vs. 3 br.	166	+0.43 [−3.03, +3.73]	0.832	< 0.001	yes	3.3 h
1 br. vs. 3 br.	173	−0.97 [−3.14, +1.29]	0.086	< 0.001	yes	2.8 h

For all three pairs the paired mean difference is within the ± 6 -hour margin, and in fact within 4.5, 3.3, and 2.8 hours (Table 4, Figure 5). Against a within-model ΔT spread of roughly 28 hours, mean between-model shifts larger than five hours are ruled out at the 90% level for every pair. This is a bound on the mean shift among onsets matched in both models, which is a stronger statement than a failure to detect a difference but a narrower one than equivalence of the

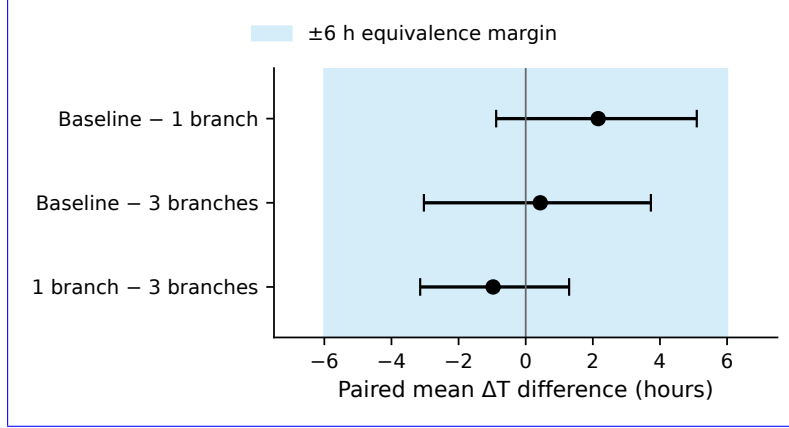


Figure 5: Onset-paired mean ΔT differences between models with 95% student-clustered bootstrap confidence intervals, against the pre-specified ± 6 -hour equivalence margin (shaded). Every interval lies well inside the margin.

distributions.

6.4. Controlling for segmentation sensitivity

To remove the **confound sample-size confound** noted in Section 6.1 we repeated the analysis using dynamic-programming segmentation (Jackson et al., 2005) with a fixed number of change points per subject, sweeping $k \in \{1, 2, 3\}$. Under this scheme every configuration produces exactly k change points for every subject.

Figure 6 illustrates the controlled analysis.

Three conclusions follow.

Sample sizes equalise With the number of detected change points held equal, the number of matched pairs becomes similar across configurations at every granularity, confirming that the variation observed under a fixed penalty was caused by (754–763 at $k = 1$, against 914–1120 under the fixed penalty; Table 5). This is consistent with the fixed-penalty differences in N arising from the penalty interacting with differently shaped confidence distributions.

Wasserstein distances shrink monotonically with granularity, from at most 3.55 h at $k = 1$ to at most 1.64 h at $k = 3$, against a distribution width of approximately 30 hours. No comparison reaches significance at any granularity.

Comparing per-repetition anticipation rates between the baseline and three-branch configurations under a fixed penalty gives $p = 0.031$, which would conventionally be read as a real difference. The pooled-distribution comparison gives $p = 0.908$ **stress trajectories**. The anticipation rates remain

Table 5: ΔT with segmentation granularity held constant. Every configuration has exactly k change points per subject; the number of matched pairs N (change points with an onset inside the window) becomes similar across configurations at every k . The anticipation rate is the repetition mean $\pm$ SD; the median ΔT is 0 hours in every cell. Time values in hours.

k	Configuration	N	Mean	Std	Anticipation
1	LSTM baseline	757	+7.48	29.2	$25.0 \pm 4.0\%$
	Personalised, 1 br.	763	+8.04	30.8	$22.9 \pm 4.3\%$
	Personalised, 3 br.	754	+6.40	30.2	$23.5 \pm 4.6\%$
2	LSTM baseline	1469	+8.35	27.1	$20.6 \pm 1.7\%$
	Personalised, 1 br.	1477	+7.06	29.3	$22.7 \pm 2.4\%$
	Personalised, 3 br.	1479	+6.91	29.8	$23.4 \pm 3.6\%$
3	LSTM baseline	2194	+7.03	27.9	$22.6 \pm 1.4\%$
	Personalised, 1 br.	2184	+7.60	29.3	$21.4 \pm 2.1\%$
	Personalised, 3 br.	2160	+7.04	29.3	$22.5 \pm 2.4\%$

within a five-point band (20.6–25.0%) with overlapping repetition spreads, and the controlled analysis gives $p \geq 0.059$ at every granularity. The apparent effect was a segmentation artifact median ΔT is 0 hours in every cell. These are descriptive summaries; the inferential comparison is the paired analysis of Section 6.3.

6.5. On the anticipation rate as a summary statistic

The disappearing effect above has a mechanical explanation that generalises beyond this dataset. The anticipation rate counts anticipation rate is the proportion of ΔT values falling below zero. Any widening of the distribution therefore increases it, whether or not the distribution has shifted earlier. It is an intuitive summary, and our data show two ways in which it can mislead when reported alone.

Under the fixed penalty, the three-branch configuration’s interquartile range was 52% First, it can differ between models whose central timing does not. The per-repetition anticipation rates of the baseline and the single-branch model differ ($22.8 \pm 2.1\%$ against $26.4 \pm 2.7\%$; two-sample t -test $p = 0.003$, and $p = 0.031$ for baseline against three branches; descriptive, since repetitions share folds). That difference in the rate is real. What it does not show is a shift in timing: all three medians are 0 hours, and the paired mean difference is bounded within 4.5 hours (Section 6.3). The personalised models’ interquartile ranges are a quarter wider than the baseline’s (48.8 against 32.0 hours) while its median moved by one hour. The distribution fanned out rather than shifting. Reporting the anticipation rate alone would have

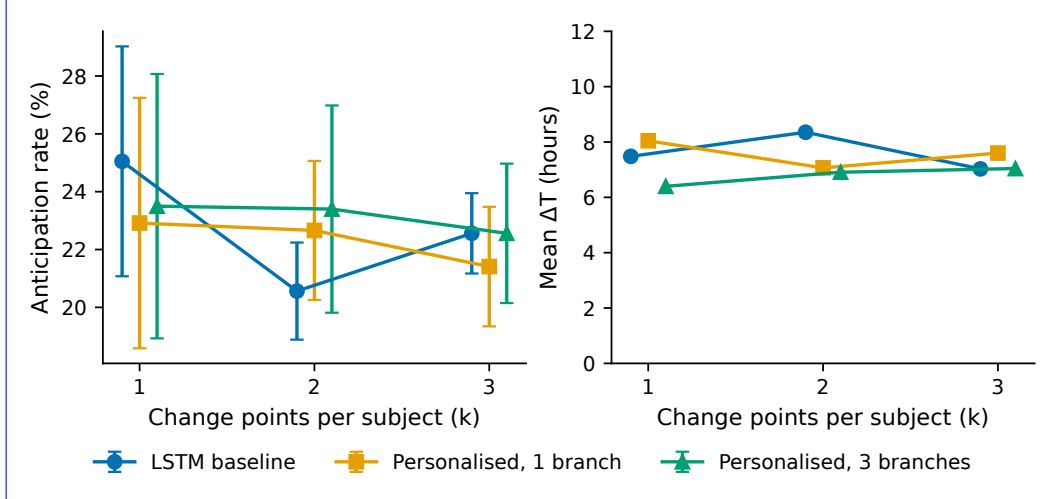


Figure 6: Invariance under controlled segmentation. Left: anticipation rates with standard deviation across repetitions, at each granularity. Right: mean ΔT (hours); the median is 0 hours for every configuration at every granularity.

characterised this as improved anticipation. It was not 30 and 29 against 24 hours), so more of their mass lies on both sides of zero. A proportion below a threshold can increase when a distribution widens without moving, although it need not: widening a distribution that is symmetric about zero leaves the proportion unchanged. A higher anticipation rate should therefore not be read as earlier detection unless the median or mean has moved with it.

Second, the rate has no meaning without a reference. The higher personalised rates are not significantly above what random change points give, and the two null models of Section 6.2 themselves differ by 13 points.

We recommend that studies using threshold-crossing summary statistics of this kind report the median, interquartile range, and a distributional distance a paired or distributional comparison, and a chance-level null alongside them.

6.6. A targeted training objective

If ΔT does not respond to architectural changes, it may respond to the training objective. We implemented a temporally asymmetric loss that up-weights examples preceding a stress onset and downweights those following it, with the explicit intent of shifting detections earlier.

Onsets. For each training subject u , examples are sorted by timestamp $\tau_{u,1} < \dots < \tau_{u,n_u}$ (hours). A training onset is any index $j \geq 2$ with $y_{u,j} > y_{u,j-1}$,

giving the onset set $\mathcal{O}_u = \{\tau_{u,j}\}$. Onsets are computed from training labels only; validation labels are never used.

Weights. For example i of subject u with timestamp τ_i , let $o_i = \arg \min_{o \in \mathcal{O}_u} |\tau_i - o|$ (if two onsets are equally near, the earlier one is used) and $g_i = \tau_i - o_i$. With asymmetry strength $\lambda \geq 0$ and a temporal window $H = 72$ h (equal to the ΔT matching window),

$$w_i = \begin{cases} 1 + \lambda, & -H \leq g_i < 0 \quad (\text{pre-onset}), \\ (1 + \lambda)^{-1}, & 0 \leq g_i \leq H \quad (\text{onset and post-onset}), \\ 1, & |g_i| > H \text{ or } \mathcal{O}_u = \emptyset. \end{cases} \quad (3)$$

$\lambda = 0$ recovers the standard objective; the pre/post weight ratio is $(1 + \lambda)^2$, i.e. 2.25 at $\lambda = 0.5$ and 4 at $\lambda = 1$. Weights are fixed before training and are not renormalised.

Objective. For a mini-batch $\mathcal{B}$, the classification term is

$$\mathcal{L}_{\text{cls}} = \frac{\beta}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_i \kappa_{y_i} \left(-\log p_i^{(y_i)} \right), \quad (4)$$

where $\kappa = (0.6456, 0.5635, 1.0000)$ are the reference implementation's class weights for the three stress classes and $\beta = 1$ is its classification-loss coefficient. The mini-batch size is 1, as in the reference implementation. The total loss is $\mathcal{L} = \alpha \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{cls}}$, where $\mathcal{L}_{\text{rec}}$ is the autoencoder reconstruction term (L1, summed over elements) with $\alpha = 10^{-4}$. The asymmetric objective was tested on the LSTM baseline configuration, in which the reconstruction term is switched off, so for these runs $\mathcal{L} = \mathcal{L}_{\text{cls}}$. The temporal weights apply to the training loss only. The epoch is selected by peak validation micro-F1, exactly as for the symmetric baseline, so the two differ only in the training objective.

The objective did not move ΔT (Table 6, Figure 7): the Mann-Whitney p values are 0.78 and 0.94, and the onset-paired differences are -0.13 and -1.78 hours with confidence intervals comfortably inside the ± 6 -hour equivalence margin. It did produce a genuine side effect: macro-F1 improved by 1.27 to 2.05 points (3.4 points at $\lambda = 0.5$ and 3.1 points at $\lambda = 1.0$ (fold-paired Wilcoxon signed-rank $p < 0.005$), because the asymmetric weighting incidentally rebalanced the class distribution and benefited the minority class $p = 0.002$ and $p = 0.010$; descriptive, since folds are shared across repetitions), at the cost of a small drop in micro-F1 (0.455 and 0.467

Table 6: ΔT under a temporally asymmetric training objective at two strengths (λ , Eq. 3). The objective did not shift the anticipation window. The Mann-Whitney test and the onset-paired mean difference (with 95% student-clustered bootstrap CI, in hours) are both against the symmetric baseline; medians are 0 hours throughout. Macro-F1 is mean $\pm$ SD over 50 folds.

Configuration	N	Mean ΔT (h)	Anticipation	MW p	Paired Δ [95% CI]	Macro-F1
Symmetric baseline	1120	+8.40	22.8%	—	—	0.332 ± 0.044
Asymmetric, $\lambda = 0.5$	1108	+8.04	24.6%	0.783	-0.13 [-2.73, +2.86]	0.366 ± 0.046
Asymmetric, $\lambda = 1.0$	1124	+8.22	23.3%	0.940	-1.78 [-5.26, +0.95]	0.363 ± 0.056

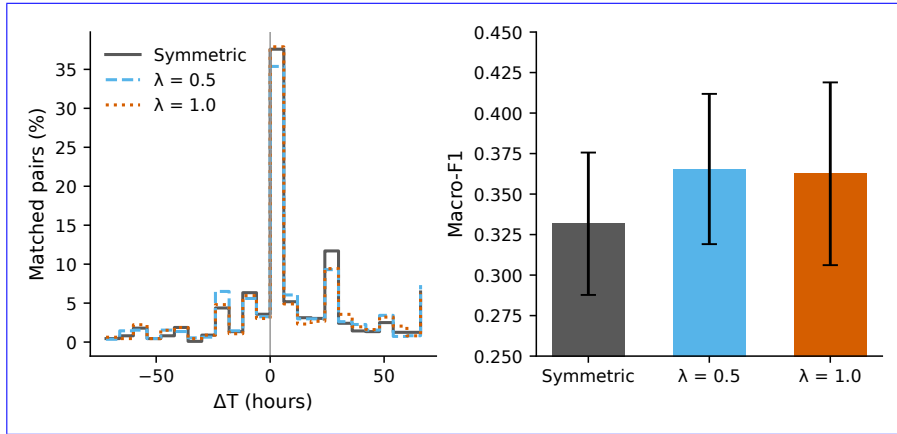


Figure 7: A temporally asymmetric objective constructed to shift ΔT earlier does not do so at either strength tested (left: the three ΔT distributions, in 6-hour bins, nearly coincide). The side effect, a macro-F1 improvement (right; error bars one SD across 50 folds), is not the intended outcome.

against 0.478). A macro-F1 gain alongside a micro-F1 loss is the pattern expected if the weighting shifted predictions toward the less frequent classes, and we interpret it that way, but we did not record per-class scores and so cannot confirm the mechanism. This is a real result, but not the one the experiment was designed to produce.

6.7. Training the models to forecast

If ΔT measures a property of the sensing modality and annotation protocol, it should behave consistently on a completely different sensing modality with a different annotation protocol. WESAD provides this test : 15 subjects wearing chest and wrist sensors (ECG, EDA, BVP, EMG, respiration, temperature, three-axis acceleration) during a controlled Trier Social Stress Test, labelled at fine temporal resolution rather than by daily EMA prompt. A natural objection to everything above is that models

trained to predict the *current* label have no reason to anticipate, so the invariance is unsurprising. We accept the premise and test the obvious remedy: retrain the models to predict the *future* label. For horizon $h \in \{1, 2, 3\}$, each training example’s label is replaced by the same student’s h -th next EMA response (the last h examples per student are dropped), and the baseline and three-branch models are retrained from scratch (5 repetitions $\times$ 5 folds per horizon). The median time gap to the target response is 16, 30, and 49 hours at $h = 1, 2, 3$. ΔT is still measured against the current-label onsets, so a model that truly anticipates by its trained horizon would push the ΔT distribution strongly negative.

We extracted ΔT from the 15 leave-one-subject-out fold records produced during the corrected WE-SAD reproduction (Section 4). Each record contains the model’s softmax probability array across three classes (baseline, stress, amusement) and the corresponding ground-truth labels for that subject’s test sequence. The class-1 (stress) probability serves as the confidence trajectory, with stress onsets defined as transitions from non-stress (baseline or amusement) into stress

Table 7: Prediction-horizon experiment. Micro-F1 is mean $\pm$ SD over 25 folds (5 repetitions); $h = 0$ is the main 50-fold run. The median gap is the median time between an example and its target label. ΔT is computed against current-label onsets; its median is 0 hours in every row. Brackets are 95% intervals from a bootstrap that resamples students. The null column is the random-change-point null (mean and 95% interval over 1,000 draws) for that row’s own validation sequences; p is the proportion of null draws at or above the observed rate, uncorrected. Rows with different h are evaluated on different validation sequences and are not directly comparable; see Table 8.

Configuration	h	Gap (h)	Micro-F1	N	Mean ΔT (h)	Anticipation (%)	Random-change-point null (%), p
LSTM baseline	0	—	0.478 ± 0.030	1120	$+8.4 [+4.3, +13.8]$	22.8 [14.9, 30.0]	26.2 [23.8, 28.8], 1.00
	1	16	0.478 ± 0.037	618	$+4.3 [+0.8, +8.6]$	28.3 [22.6, 33.3]	29.3 [26.1, 32.5], 0.70
	2	30	0.482 ± 0.032	541	$+3.6 [+0.3, +7.7]$	32.7 [22.5, 40.5]	28.9 [25.4, 32.4], 0.02
	3	49	0.490 ± 0.030	532	$+3.3 [-0.7, +7.7]$	34.2 [26.7, 40.3]	31.5 [27.8, 35.4], 0.08
Personalised, 3 br.	0	—	0.611 ± 0.022	914	$+6.7 [+1.4, +12.4]$	26.6 [18.3, 34.2]	25.9 [23.3, 28.6], 0.32
	1	16	0.609 ± 0.019	526	$+3.8 [+0.2, +8.4]$	32.1 [25.1, 38.5]	28.2 [24.9, 31.8], 0.01
	2	30	0.610 ± 0.014	478	$+1.3 [-2.5, +5.1]$	33.3 [26.0, 40.7]	29.8 [26.0, 33.7], 0.04
	3	49	0.626 ± 0.021	494	$+0.6 [-3.3, +4.4]$	30.6 [23.2, 37.4]	31.0 [27.3, 34.8], 0.57

Three results follow (Tables 7 and 8, Figure 8).

WESAD sequences are short: approximately 60 windows per fold, against StudentLife’s hundreds of hours per subject. The PELT penalty of 1.0 used in Section 5 is too strict for this shorter regime, admitting only 16 change points across all 15 folds. First, against the expectation that forecasting is harder than nowcasting, micro-F1 does not decay with horizon. It is flat to slightly rising (0.478 to 0.490 for the baseline; 0.611 to 0.626 for the three-branch model at a 49-hour median gap). A calibration sweep over $\{0.05, 0.1, 0.5, 1.0, 2.0\}$ identified $\rho = 0.5$ as the operating point that admits a healthy sample size (66 matched pairs) while remaining stable relative to lower penalties. Table 6 reports the sweep. Self-reported stress is autocorrelated enough

Table 8: Forecast-trained against present-state models on identical validation sequences. For each forecast run, the present-state ($h = 0$) model’s out-of-fold predictions for the same examples are assembled into the same sequences and analysed in the same way. The paired difference is forecast minus present, per onset matched in both (median across repetitions), with a 95% interval from a bootstrap that resamples students; the last column states whether the 90% interval lies inside the ± 6 -hour margin.

Configuration	h	Mean ΔT (h)		Anticipation (%)		Onsets	Paired difference (h) [95% CI]	Within ± 6 h
		Forecast	Present	Forecast	Present			
LSTM baseline	1	+4.3	+3.4	28.3	29.1	165	+1.1 [−2.2, +4.1]	yes
	2	+3.6	+4.0	32.7	33.2	170	+2.0 [−5.2, +8.0]	no
	3	+3.3	+0.7	34.2	36.4	160	−0.6 [−5.7, +3.4]	yes
Personalised, 3 br.	1	+3.8	+1.6	32.1	35.4	171	+1.9 [−1.9, +5.9]	yes
	2	+1.3	+4.0	33.3	30.9	164	−1.0 [−5.5, +3.4]	yes
	3	+0.6	+3.8	30.6	31.1	157	+1.0 [−3.3, +5.1]	yes

that predicting the label two days ahead is as easy as predicting today’s, which suggests the models track a slow-moving person-level signal more than transition dynamics.

$N = 66$ matched pairs; mean $\Delta T = +0.30$ steps; median $\Delta T = 0$ steps; anticipation rate = 33.3%.

Second, a direct comparison of the horizon runs with the $h = 0$ runs is misleading, and we report it because it misled us. Read down Table 7, the mean ΔT falls by 3 to 6 hours and the anticipation rate rises by 4 to 11 points as the horizon grows, in all six runs, and with students resampled jointly the 95% interval for the fall in mean ΔT excludes zero in five of them. But the horizon runs are validated on different sequences from the $h = 0$ runs: the last examples of each student are dropped and the folds are stratified on the shifted labels. The random-change-point null for those sequences already shows the problem, since its anticipation rate rises by 2 to 5 points over the same horizons with no model involved.

Third, on identical sequences the difference disappears. Every example has an out-of-fold prediction from the present-state model of the same repetition, so for each validation sequence of a forecast run we rebuilt the present-state model’s confidence trajectory shifts at almost exactly the moment stress is labelled. trajectory on exactly the same examples and compared the two onset by onset (Table 8). The paired mean differences are between -1.0 and $+2.0$ hours, with no consistent sign, and every 95% interval includes zero. The 90% interval lies inside the ± 6 -hour margin in five of the six runs; the exception is the baseline at $h = 2$, whose interval reaches $+7.1$ hours. The anticipation rates of the two models on the same sequences differ by at most 3.2 points, again with every interval including zero, and each forecast run’s mean ΔT lies inside the random-change-point null interval for its own sequences. The present-state model, placed on the forecast

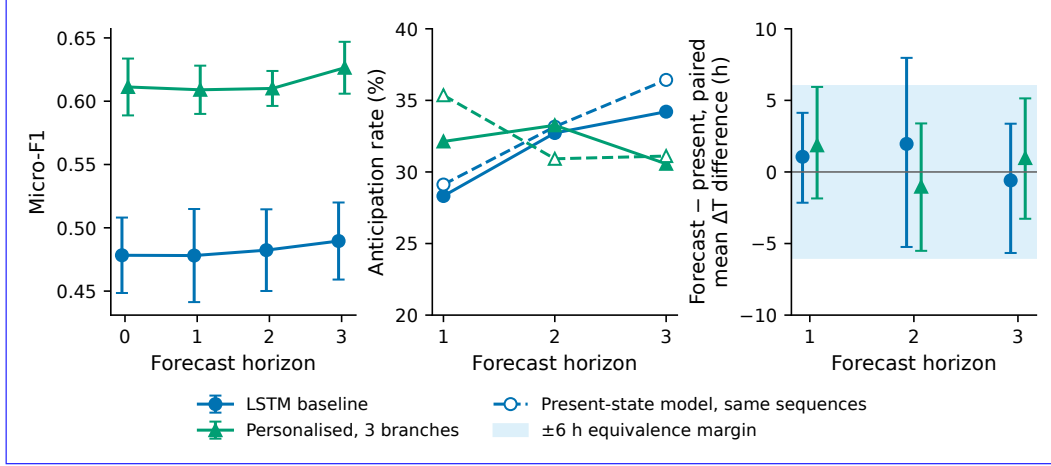


Figure 8: Prediction-horizon experiment. Left: micro-F1 (error bars one SD across folds). Centre: anticipation rate of the forecast-trained model (solid) and of the present-state model on the same validation sequences (dashed, open markers). Right: onset-paired mean ΔT difference, forecast minus present, with 95% student-bootstrap intervals against the ± 6 -hour margin. The horizon is the number of EMA responses ahead that the model was trained to predict.

runs' sequences, shows the same lower mean ΔT and higher anticipation rate that the forecast-trained model does.

Table 7 places the WESAD result alongside the three StudentLife configurations. Training the models to forecast up to 49 hours ahead therefore did not move their timing. What moved the statistics was the composition of the evaluation sequences, which is a caution in its own right: ΔT summaries from different validation sequences are not comparable, even for the same students and the same model, and comparisons should be made on identical sequences, as the main analysis does. One caveat applies to the same-sequence comparison: within a sequence, the present-state predictions come from the five fold-models of one repetition, each out-of-fold for its own examples, and not from a single model.

6.8. Sensitivity to the penalty and the matching window

Four configurations span two sensing modalities (behavioural smartphone data and physiological wearable data), two annotation protocols (daily EMA and continuous laboratory labelling), and two dataset scales (10-week naturalistic and 2-hour controlled). The anticipation rate spans a narrow range of 24.8–33.3%. This is not a demonstration of a universal constant, but it is evidence that the ΔT framework produces stable and comparable measurements across substantially different sensing regimes. All results above fix the PELT penalty at 1.0 and the matching window at 72 hours. Figure 9

repeats the main analysis over the full grid $\text{penalty} \in \{0.5, 1, 2\} \times \text{window} \in \{24, 48, 72, 96\}$ hours. The anticipation rate stays within 21–28% in every cell with $\text{penalty} \leq 1$, rises mechanically with the window (more distant matches admit more negatives and positives alike), and is noisier at penalty 2, where the three-branch model retains as few as 152 pairs. At penalty 5 fewer than a dozen pairs survive per configuration and no estimate is meaningful; those cells are listed in Appendix B for completeness. The median ΔT is 0 hours in every cell of the grid, and the ordering of the three configurations is not stable across cells — no choice of penalty or window produces a personalisation advantage in timing.

The result should be held with appropriate caution given the WESAD sample size (66 matched pairs from 15 subjects). We report it as supporting cross-modal evidence rather than as an independent confirmation of the invariance claim.

7. Discussion

7.1. Interpretation

Four independent perturbations interventions were applied: changing the architecture, adding single-branch personalisation, adding three-branch personalisation, and changing the training objective a timing-targeted training objective, and retraining the models to forecast one to three EMA responses ahead. All four left ΔT statistically unchanged the median ΔT at 0 hours. For the two personalisation variants and the timing-targeted objective, the paired mean ΔT difference from the baseline is within the ± 6 -hour margin under a student-level bootstrap. For the forecasting runs, compared with the present-state models on identical sequences, the paired mean difference is between -1.0 and $+2.0$ hours with every interval including zero, and within the margin in five of six runs (the baseline at $h = 2$ is the exception, so equivalence is not established there). The null models show that none of the main configurations has significantly more anticipation than chance placement of change points, and that no forecasting run does after correction for multiple tests; the baseline does show onset alignment above the random-change-point null, so its change points are not unrelated to the onsets, but they fall at or after them.

The most direct reading is that ΔT measures a property of the data and the annotation protocol rather than a property of the model. The gap between when behavioural sensor signals shift and when a person reports feeling stressed appears to be set by the sensing modality and the ecological momentary assessment schedule, not by the model applied to them.

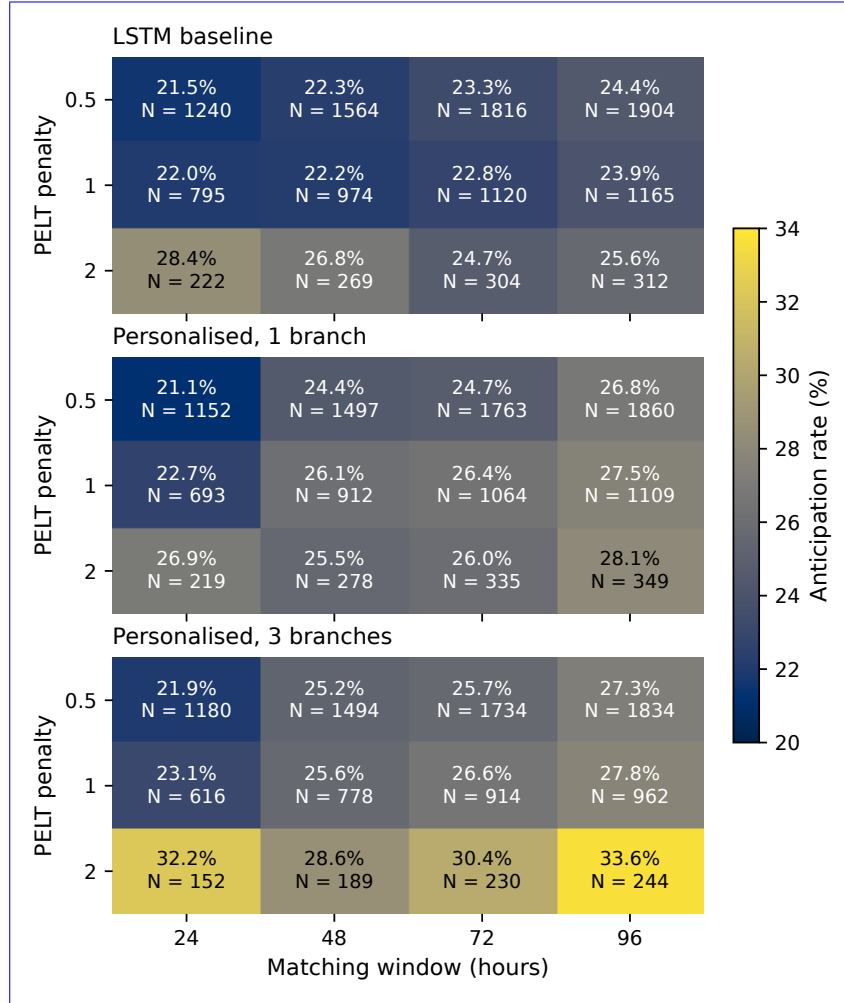


Figure 9: Anticipation rate over the PELT penalty $\times$ matching window grid, with the number of matched pairs in each cell. The median ΔT is 0 hours in every cell. Penalty 5 (not shown) leaves fewer than a dozen pairs per configuration.

This reading should be held with appropriate caution. What we demonstrate is a **bounded null** across a specific family of models on a single dataset. A null result is a statement that no effect was detected at the available statistical power, not a demonstration that no effect exists: the paired analysis rules out between-model shifts larger than 4.5 hours, but smaller shifts remain possible, and other model families might behave differently.

7.2. Implications for system design

For anyone building a stress-monitoring system intended to support intervention, the implication is direct. In this dataset roughly a quarter of stress transitions carry a detectable precursor, and that proportion did not change regardless of what was done to the model: the anticipation rate sits between 23% and 27% for the baseline, both personalised models, and both strengths of the timing-targeted objective, and the null models show that this is not significantly more than chance placement of change points produces under the same matching rule. We find no evidence that any of these systems provides early warning. Improving the classifier by 13.4 13.3 points of micro-F1 and retargeting the loss produced no measurable improvement in timing. shift in mean or median timing. Retraining to forecast up to 49 hours ahead also left timing where it was, once the forecast-trained and present-state models were compared on the same sequences.

This should be surprising to anyone who assumes accuracy and timeliness improve together, and it is invisible to standard evaluation. We suggest that studies reporting stress-prediction systems intended for intervention include a timing measurement alongside classification metrics. The ΔT framework makes this possible without changes to the model or additional annotation.

7.3. Limitations

Shared fold assignments. The ten repetitions use identical deterministic fold assignments. The same subjects, onsets, and trajectories appear in every repetition, with only model initialisation differing. This tests whether architecture moves ΔT with the data held constant. It does not establish cohort-level generality, and the effective number of independent observations is smaller than the raw sample sizes suggest.

With a distribution width of approximately 30 hours and the effective sample sizes above, this study can detect shifts on the order of several hours. A meaningful improvement in anticipation might be two or three hours, which

Statistical power. The paired equivalence analysis makes the power explicit: between-model shifts are bounded within 4.5, 3.3, and 2.8 hours for the three pairs at the 90% level. A real improvement smaller than these margins — say two hours — would not be reliably detected. The result should be read as a bounded null rather than a demonstration of no effect.

Single cohort for the invariance test. The invariance test proper derives from 23 students over one academic term (StudentLife). Each subject contributes between zero and three matched pairs per fold.

The cross-modal extension yields 66 matched pairs from 15 subjects over approximately two hours each. This is a smaller sample than the StudentLife invariance test and covers a much shorter time span. We report the WESAD result as supporting cross-modal evidence, not as an independent replication of the invariance claim

Retrospective change points. PELT and the dynamic-programming segmentation are offline procedures: each change point is located using the subject's whole validation trajectory, including examples after it. A negative ΔT therefore means that the model's trajectory, viewed in retrospect, changed before the reported onset. It does not mean that a deployed system could have raised an alert at that time, since an online detector needs evidence after the change to declare it. ΔT as measured here is at best an optimistic estimate of what an online version could achieve, and measuring real-time warning lead would require an online detector with an explicit detection delay.

The PELT penalty was calibrated separately for each dataset (1.0 for StudentLife, 0.5 for WESAD) to account for their different sequence lengths and confidence dynamics. This is standard practice for change-point detection but means the two anticipation rates are not directly numerically comparable at the level of exact percentages, only in terms of order of magnitude

Dependence on the evaluation sequence. The ΔT summaries depend on which examples make up a validation sequence. The same present-state models give a mean ΔT of +6.7 to +8.4 hours on their own folds and +0.7 to +4.0 hours on the forecast runs' folds (Section 6.7). Every comparison we draw conclusions from is made on identical sequences, but absolute values of the mean ΔT and of the anticipation rate should not be compared across studies, or across splits, without that control. In the same-sequence comparison for the forecasting runs, the present-state predictions within a sequence come from five fold-models and not one, the runs have five repetitions each, and the p values in Table 7 are uncorrected for six tests.

Heuristic matching rule. The 72-hour (StudentLife) and 10-step (WESAD) matching windows matching window and nearest-neighbour rule were selected for this study. Different choices would produce different ΔT distributions; Section 6.8 shows how far the summary statistics move across four windows and three penalties.

Temporal resolution. ΔT is computed on each student’s validation subsequence within a fold, about ten EMA responses spread over the term, so consecutive points are typically a day or more apart. A third of matched pairs have $\Delta T = 0$ exactly because the change point falls on the onset example itself. The metric cannot resolve lead or lag finer than the spacing between responses, and a denser evaluation sequence (for example a chronological hold-out) would give a sharper measurement.

Withdrawn cross-modal analysis. The originally submitted version included a ΔT analysis of the WESAD wearable dataset. While preparing this revision we found that the reference WESAD pipeline evaluates with a shuffled data loader and stores no window index, so the trajectories analysed there were not in temporal order and the reported WESAD ΔT values were not valid timing measurements. We have withdrawn that analysis. All claims in this paper now rest on StudentLife alone, and whether the result holds for physiological wearable sensing is an open question.

7.4. Future directions

The immediate priority is a larger cross-modal validation, either on a longer physiological dataset than WESAD or on additional participants under a similar controlled protocol test on physiological wearable data, with the evaluation sequence kept in temporal order. Beyond that, extending ΔT to a dataset with independent subject cohorts would address the shared-fold limitation directly. Finally, the structural problem here — detecting a latent state transition before it becomes externally observable — is not specific to stress; the same measurement applies to any domain where an early warning system is the goal. A companion paper transfers the framework to mobile network signal quality prediction, providing early evidence that the finding generalises beyond health sensing.

8. Conclusion

We introduced ΔT , a measurement of when a stress-prediction model detects a transition relative to when the person reports it, and showed that it its central tendency does not respond to changes in model architecture, to the addition

of personalisation , or to a training objective designed specifically to move it, and that retraining the models to forecast up to 49 hours ahead changes neither their accuracy nor, on identical validation sequences, their timing. Across three configurations spanning 13.4 13.3 points of micro-F1 on StudentLife, the ΔT distributions are statistically indistinguishable, and the result survives a control for segmentation sensitivity that eliminates an apparent effect present in the uncontrolled analysis. Extension to WESAD places the anticipation rate within a narrow band of 24.8–33.3% across four configurations, two sensing modalities (behavioural and physiological), two annotation protocols (daily EMA and continuous laboratory labelling), and two dataset scales. Paired mean ΔT differences on onsets matched in both models fall within a ± 6 -hour equivalence margin, no configuration shows significantly more anticipation than either of two chance-level null models, and the median ΔT is 0 hours under every segmentation setting we tried. These are retrospective alignment measurements, not real-time warning leads.

Timing and accuracy are separable properties of a stress-prediction model. Standard evaluation metrics measure the former accuracy and are silent about the latter timing. Systems intended to anticipate stress rather than merely recognise it will require methods that target timing directly, and evaluation protocols that measure whether they succeed.

CRedit authorship contribution statement

David Olutunde Daniel: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The research was conducted during a research internship at AT&T, funded through Tech Mahindra (Americas) Inc.

Ethics statement

This study is a secondary computational analysis of publicly available, de-identified datasets collected and released by their original investigators. No new data collection involving human participants was undertaken, and no personally identifying information was accessed. Additional institutional review board approval was therefore not required.

Data availability

The StudentLife dataset is publicly available from Dartmouth College. The WESAD dataset is publicly available from the University of Siegen (Schmidt and Reiss, 2018). The reference implementation and its preprocessed data assets are distributed through the public repository cited in the references (Information Fusion Lab, 2021). Analysis scripts supporting the ΔT framework, the controlled-segmentation robustness check tables of matched pairs underlying every result, and the figures in this paper are available from the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used a generative AI assistant solely to improve the clarity, readability, and language of the manuscript text. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. All experimental design, code, data processing, numerical results, figures, and scientific conclusions were produced and verified independently by the author. No AI-generated text was included without human review.

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Appendix A. Sensitivity to the trajectory definition

Table A.9 repeats the main analysis under three trajectory definitions at $\text{jump} = 1$ — the expected stress level of Eq. 2 (main text), the maximum class probability used in the originally submitted version, and the high-stress class probability $p_t^{(2)}$ — and, for completeness, under the originally submitted configuration (maximum probability with the ruptures default candidate grid of every fifth

sample). These are descriptive sensitivity results; no equivalence test was run on them. The pattern is the same under every definition: anticipation rates stay within 22–27%, and the medians are 0 hours for every $\text{jump} = 1$ analysis. The submitted configuration shows how the coarse candidate grid shrinks the number of matched pairs by roughly a factor of three and inflates the medians to +4.0 to +5.5 hours, which accounts for the differences between the originally submitted tables and those in this revision.

Table A.9: ΔT under alternative trajectory definitions. Anticipation is the repetition mean $\pm$ SD. All analyses use PELT with penalty 1.0 and the 72-hour window.

Trajectory	Configuration	N	Mean (h)	Median (h)	Anticipation
Expected stress, s_t (main text)	LSTM baseline	1120	+8.40	0	$22.8 \pm 2.1\%$
	Personalised, 1 br.	1064	+6.97	0	$26.4 \pm 2.7\%$
	Personalised, 3 br.	914	+6.65	0	$26.7 \pm 4.8\%$
Maximum probability	LSTM baseline	965	+6.81	0	$24.0 \pm 4.4\%$
	Personalised, 1 br.	1073	+7.71	0	$25.9 \pm 3.8\%$
	Personalised, 3 br.	946	+8.69	0	$24.1 \pm 5.4\%$
$P(\text{high stress}), p_t^{(2)}$	LSTM baseline	1056	+10.32	0	$22.2 \pm 2.8\%$
	Personalised, 1 br.	1092	+7.09	0	$25.8 \pm 2.0\%$
	Personalised, 3 br.	932	+8.06	0	$25.8 \pm 2.2\%$
Max. probability, default grid (as submitted)	LSTM baseline	332	+11.05	+4.0	$24.7 \pm 5.8\%$
	Personalised, 1 br.	420	+13.69	+5.5	$24.7 \pm 3.5\%$
	Personalised, 3 br.	358	+15.78	+5.5	$23.7 \pm 4.0\%$

Appendix B. Full penalty and matching-window grid

Table B.10 lists every cell of the sensitivity grid of Section 6.8, including penalty 5, where too few matched pairs remain for the rates to be meaningful. The median ΔT is 0 hours in every cell.

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Table B.10: Anticipation rate, with the number of matched pairs in parentheses, for every PELT penalty and matching window.

Penalty	Window (h)	LSTM baseline	Personalised, 1 br.	Personalised, 3 br.
0.5	24	21.5% (1240)	21.1% (1152)	21.9% (1180)
	48	22.3% (1564)	24.4% (1497)	25.2% (1494)
	72	23.3% (1816)	24.7% (1763)	25.7% (1734)
	96	24.4% (1904)	26.8% (1860)	27.3% (1834)
1	24	22.0% (795)	22.7% (693)	23.1% (616)
	48	22.2% (974)	26.1% (912)	25.6% (778)
	72	22.8% (1120)	26.4% (1064)	26.6% (914)
	96	23.9% (1165)	27.5% (1109)	27.8% (962)
2	24	28.4% (222)	26.9% (219)	32.2% (152)
	48	26.8% (269)	25.5% (278)	28.6% (189)
	72	24.7% (304)	26.0% (335)	30.4% (230)
	96	25.6% (312)	28.1% (349)	33.6% (244)
5	24	30.0% (10)	25.0% (8)	33.3% (6)
	48	30.0% (10)	25.0% (8)	25.0% (8)
	72	30.0% (10)	20.0% (10)	18.2% (11)
	96	30.0% (10)	20.0% (10)	18.2% (11)

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